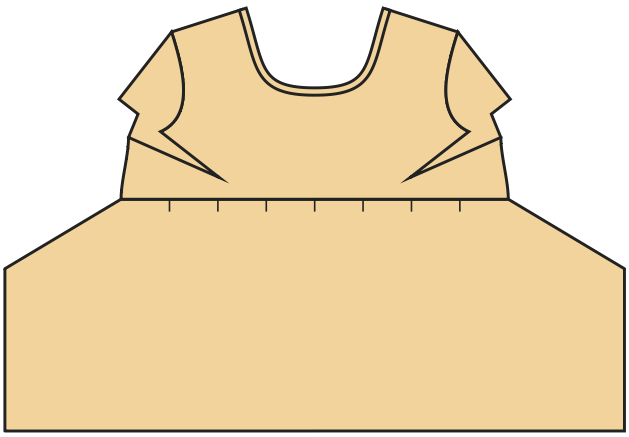
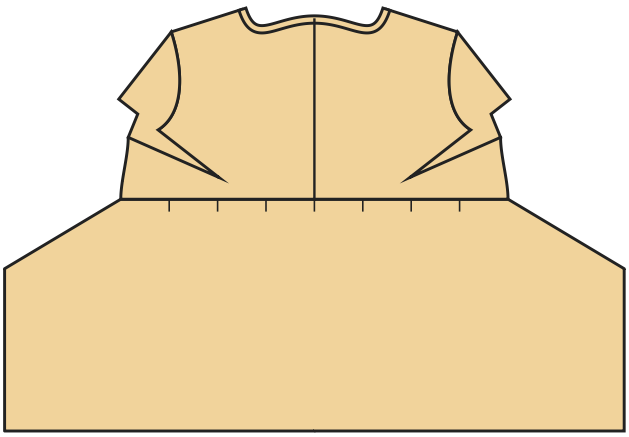
Dress with gathered skirt

doll 45 cm (made to measure) · wide fit, Woven (0 % stretch)

6 parts · 24 steps · advanced · about 3.5 h



Front view



Back view

Sewn with: Sewing machine



Sewing machine



Iron

Read the steps once before you start. Every picture is drawn from your pattern pieces. Sew a test garment from cheap fabric first and adjust before you cut the good fabric.

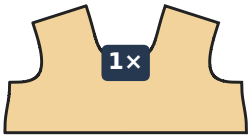
Parts



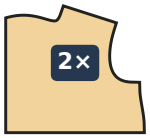
E · Skirt front (cut on fold)



F · Skirt back



A · Front (cut on fold)



B · Back



C · Sleeve



D · Neck binding (bias)

Seam allowance: included: 0.6 cm · hem 0.6 cm, 1.0 cm

Material

Main fabric
0.25 m at 150 cm width

Sewing machine
straight stitch 2.5 mm; wide zigzag 4 × 1.5 mm over the edge; straight stitch 3 mm; 2 rows of the longest straight stitch, loose ends

Polyester sewing thread
In the colour of the fabric.

Chalk or a fabric pen
For notches, labels and placement lines.

Velcro
The back opens completely: close it with narrow Velcro (1 cm) or 2–3 small snaps.

Iron
steam, cotton setting

Pins or clips
Pin across the seam line.

Colours in this booklet

Main fabric
right side · wrong side

Sew now (machine)

Sew by hand

Fold / press line

Marking (chalk)

Pin

Notch: match them

Right/wrong side of the fabric = the side of the fabric. Left/right (as worn) = seen from the child wearing it: the left front leg is on the right when you look at the child.

Basics: Sewing machine

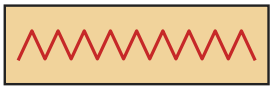
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

Basics: the sewing machine



Straight stitch

Seams in woven fabric, 2.5 mm; backstitch at the start and the end.



Zigzag

Knit seams (narrow) and finishing the edges (wide).

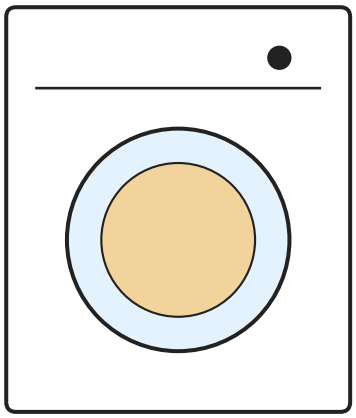


Topstitching

Right side up, with the same distance to the edge all the way; use the edge of the presser foot as a guide.

1 Pre-wash the fabric

1. Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.



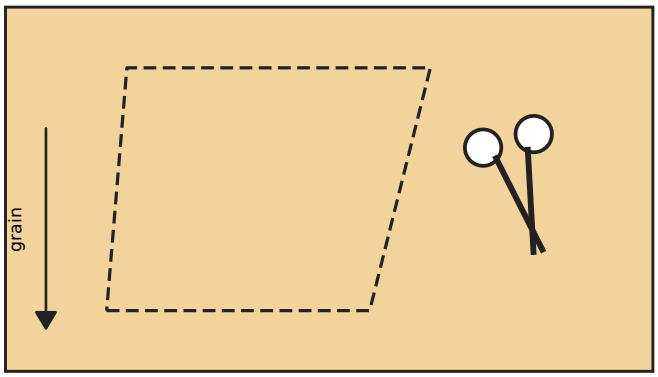
30–40°

2 Cut out all pieces

Scissors

1. Follow the cutting layout
2. Grainline arrows parallel to the selvage.

Tip: Cut both layers at once with long strokes; keep the fabric flat on the table.

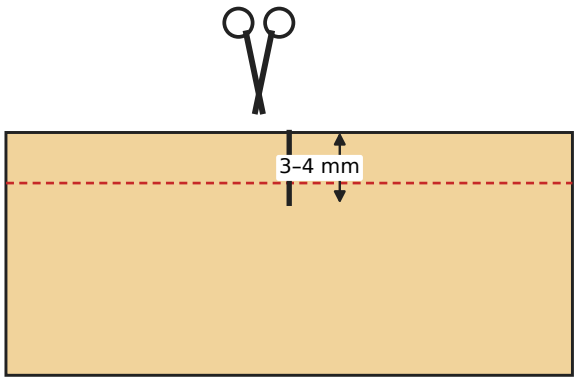


3 Snip the notches

Chalk

1. Cut 3–4 mm into the seam allowance at every notch.

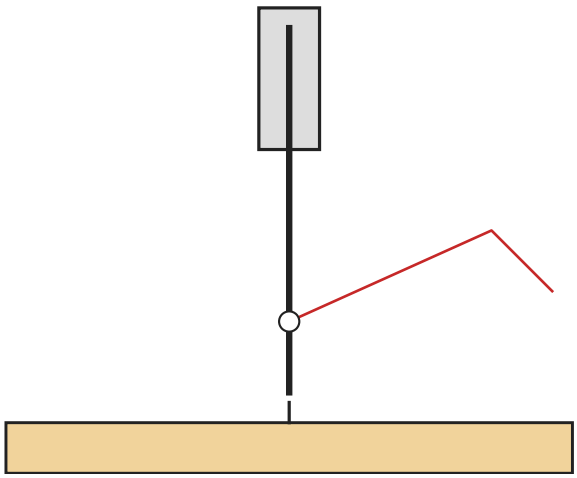
⚠ Important: Never deeper than half the seam allowance.



4 Set up the machine

1. Universal needle 80, polyester thread.
2. Sew a test seam on a scrap first.

Tip: See the basics page at the front of the booklet.



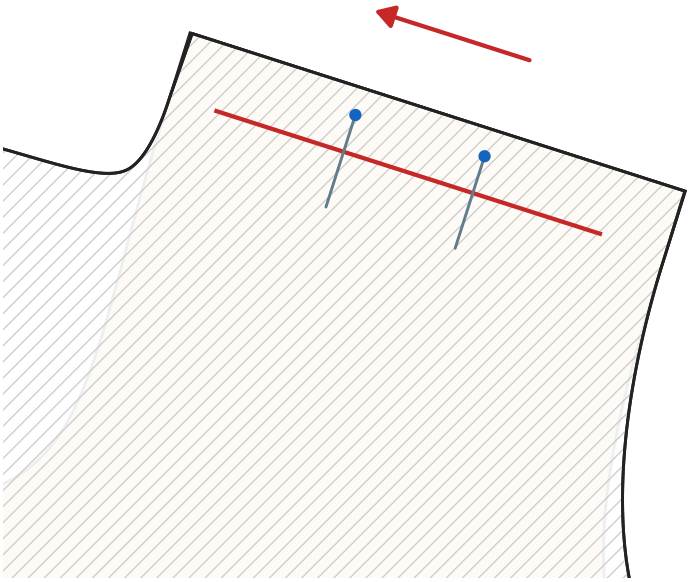
5 Sew the shoulder seams

Sewing machine

Pins

1. Front and back right sides together.

Sewing machine: straight stitch 2.5 mm



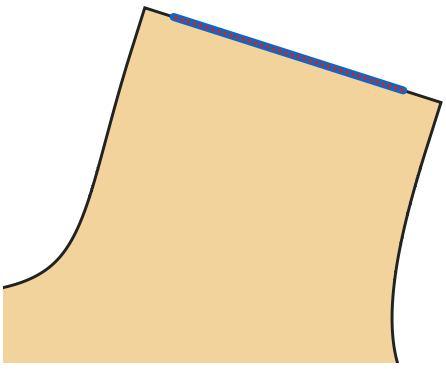
2×

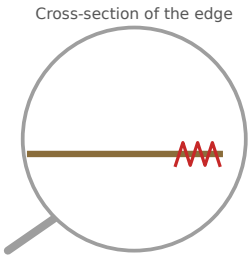
6 Finish the shoulder seams

Sewing machine

1. Both allowances together
2. Press them towards the back.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge





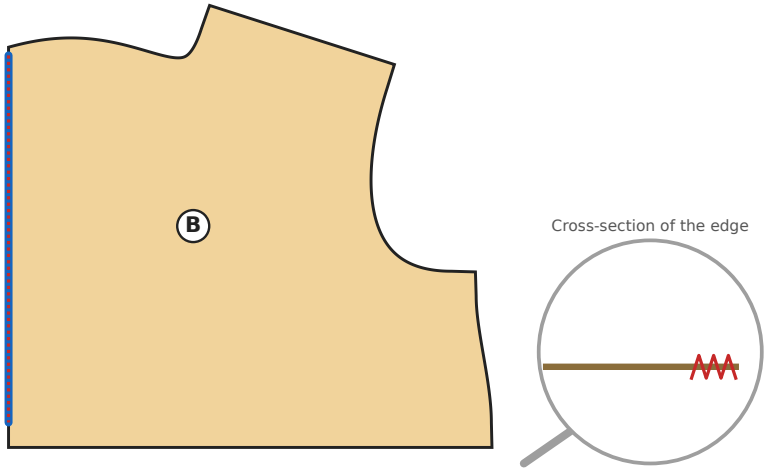
2×

7 Finish the opening edges

Sewing machine

1. Both centre back edges.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge



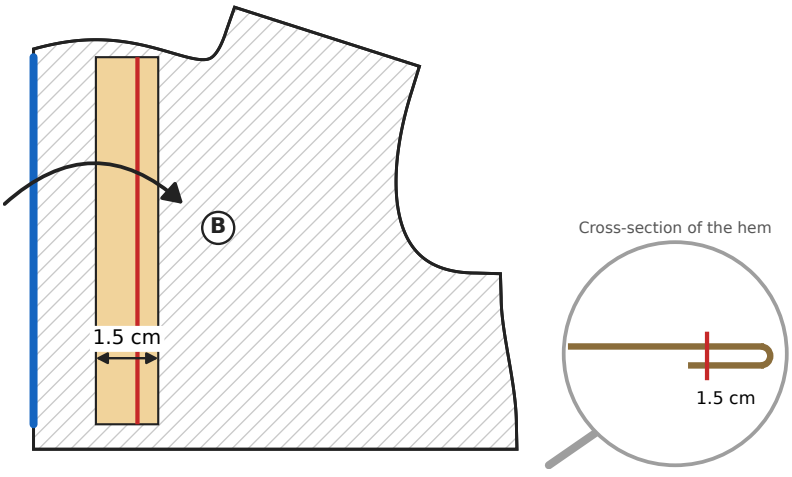
2×

8 Turn in the opening edges

Sewing machine

1. Press 1.5 cm to the inside and stitch it down.

Sewing machine: straight stitch 3 mm



2×

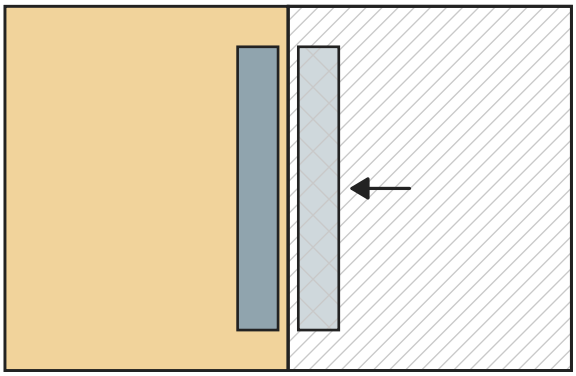
9 Sew on the Velcro

Sewing machine

1. The edges overlap at the centre back: soft side on the left back, hook side on the right back (or 2–3 snaps).

Sewing machine: straight stitch 3 mm

⚠ Important: The garment opens completely: easy dressing.

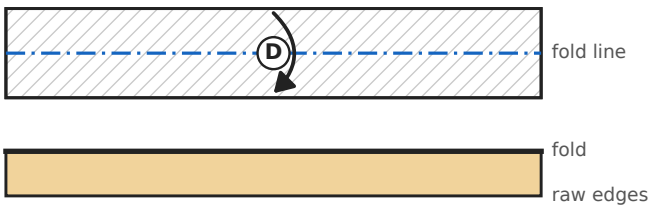


10 Fold the binding in half

Iron

1. Lengthwise, wrong sides together
2. Press.

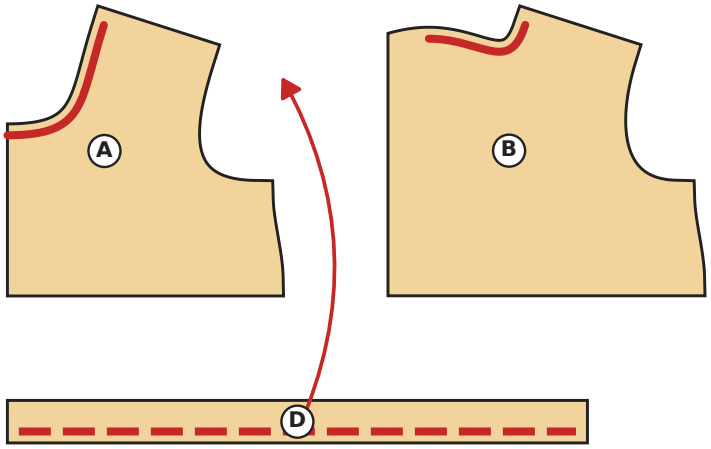
Iron: steam, cotton setting



11 Sew the neck binding on

Sewing machine

Pins

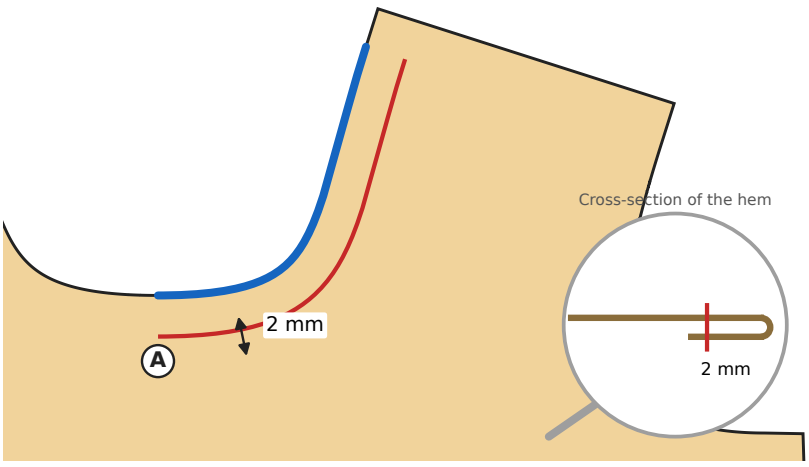


1. Raw edges to the neckline, right sides together
2. Let each end stick out past the opening edge by the allowance.

Sewing machine: straight stitch 2.5 mm

12 Stitch the binding down

Sewing machine



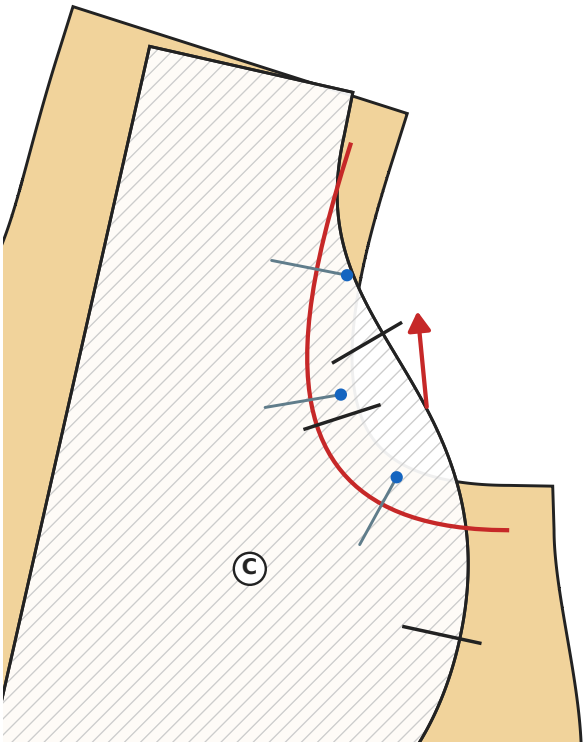
1. Fold the ends in, turn the binding to the inside over the allowance and stitch close to the fold.

Sewing machine: straight stitch 3 mm

13 Set in the sleeves

Sewing machine

Pins



1. Top notch of the sleeve cap on the shoulder seam, front/back notches meet
2. Sew the sleeve in flat.

Sewing machine: straight stitch 2.5 mm

⚠ Important: Sleeve on the bottom: the machine eases the cap in.

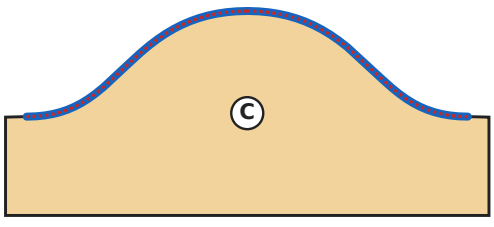
2×

14 Finish the armhole seams

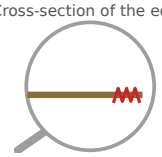
Sewing machine

1. Both allowances together
2. Press them towards the back.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge



Cross-section of the edge



2×

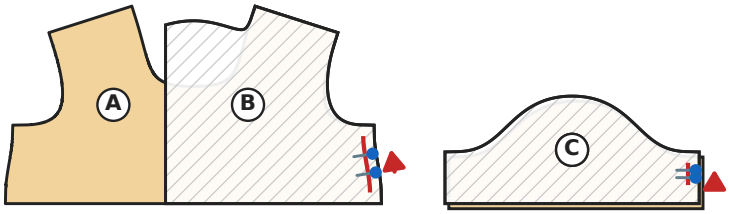
15 Sew side and sleeve seams

Sewing machine

Pins

1. In one go from the hem over the underarm to the sleeve hem
2. The underarm seams meet exactly.

Sewing machine: straight stitch 2.5 mm



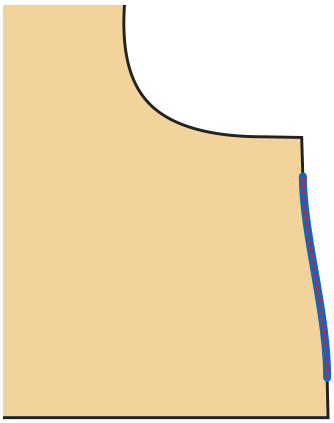
2×

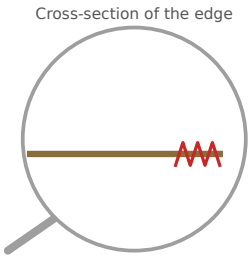
16 Finish the side seams

Sewing machine

1. Both allowances together
2. Press them towards the back.

Sewing machine: wide zigzag 4 × 1.5 mm over the edge





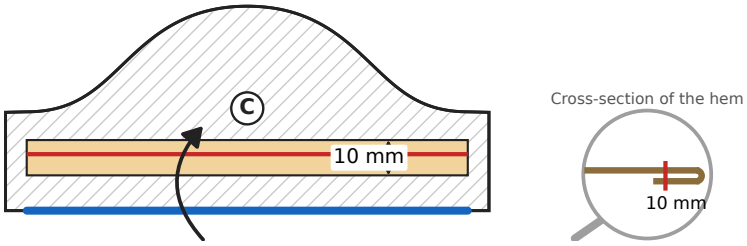
2×

17 Hem the sleeves

Sewing machine

1. Press 10 mm to the inside.

Sewing machine: straight stitch 3 mm



2×

18 Sew the skirt centre back seam

Sewing machine

Pins

1. From the hem up to the notch
2. Above the notch it stays open (continues the back opening).

Sewing machine: straight stitch 2.5 mm



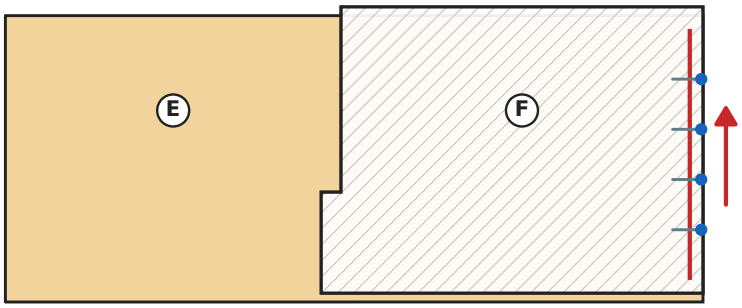
19 Sew the skirt side seams

Sewing machine

Pins

1. Skirt front and back right sides together.

Sewing machine: straight stitch 2.5 mm

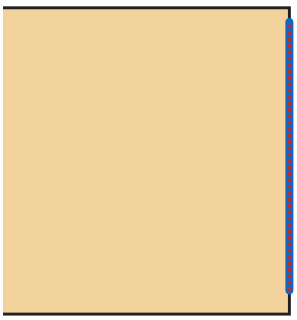


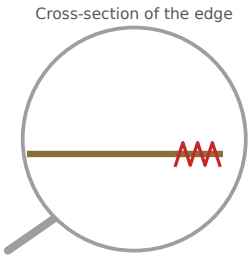
2×

20 Finish the skirt side seams

Sewing machine

Sewing machine: wide zigzag 4 × 1.5 mm over the edge





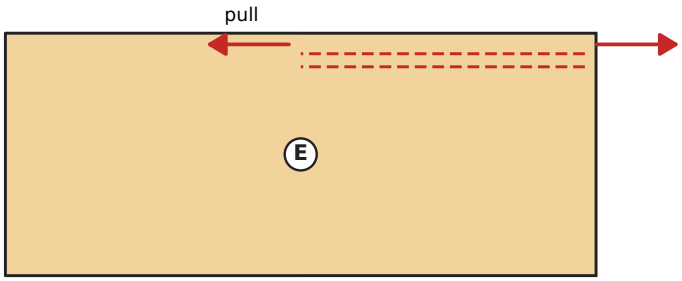
2×

21 Gather the skirt top edge

Sewing machine

1. Two rows of long stitches
2. Pull them up to the bodice waist (gather ratio 1.6), spread the gathers evenly.

Sewing machine: 2 rows of the longest straight stitch, loose ends

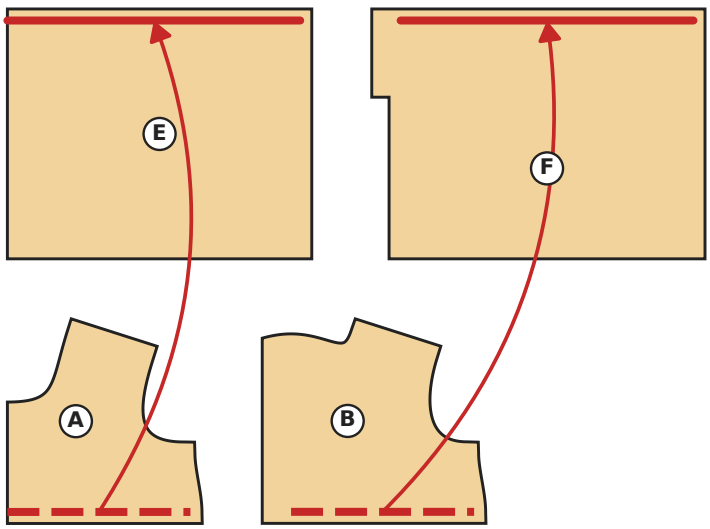


22 Sew skirt and bodice together

Sewing machine · Pins

1. Skirt right sides together to the bodice: side seams meet, centre front/back meet.

Sewing machine: straight stitch 2.5 mm

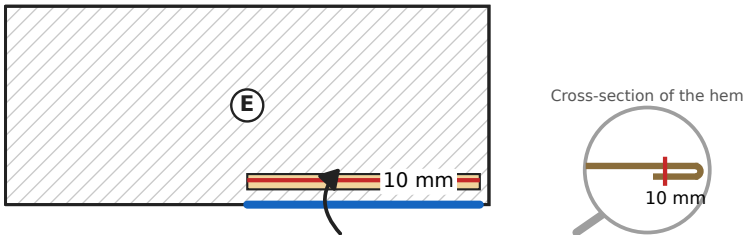


23 Sew the hem

Sewing machine

1. Press the allowance under twice and stitch.

Sewing machine: straight stitch 3 mm



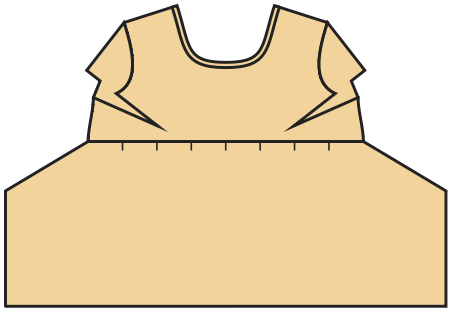
24 Done!

1. Press all seams.

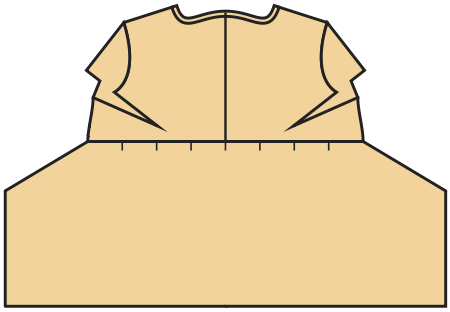


Done!

After the test garment: Try the test garment on the child: length, width at the hip and the waistband. Adjust the measurements or options and print the pattern again before you cut the good fabric.



Front view



Back view

Dress with gathered skirt · doll 45 cm (made to measure)

wide fit, Woven (0 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

Files

- PDF A4 tiles to glue (3 pages + overview): `doll_dress_45cm_a4_tiled.pdf`
- Preview A4 tiles: `doll_dress_45cm_a4_preview.png`
- SVG master 1:1: `doll_dress_45cm_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

Body measurements

Body measurement	Value	Source
Body height	45.0 cm	user
Chest	30.0 cm	user
Waist	28.0 cm	user
Hip	31.0 cm	user
Head	32.0 cm	user
Neck	15.1 cm	derived
Inseam	14.0 cm	user
Side length	22.0 cm	user
Arm length	12.0 cm	user
Back length	9.0 cm	derived
Shoulder width	3.2 cm	derived
Thigh	18.0 cm	derived
Ankle	11.4 cm	derived

Finished measurements (virtually sewn)

Finished measurement	Value
Chest width	36.9 cm
Hem width	60.9 cm
Dress length from neck point	20.7 cm
Sleeve length	4.2 cm
Sleeve opening	12.5 cm
Upper-arm width	12.5 cm
Neckline circumference	20.4 cm
Waist seam (bodice)	38.1 cm
Skirt top edge (ungathered)	60.9 cm
Skirt length	11.4 cm

Pattern pieces

Piece	Cutting	Fabric	Seam allowance
Front	cut 1 on fold	Main fabric	seam allowance 0.6 cm included; Neckline: 0.4 cm
Back	cut 2 (mirrored pair)	Main fabric	seam allowance 0.6 cm included; Back opening (turn under): 1.5 cm; Neckline: 0.4 cm
Sleeve	cut 2 (mirrored pair)	Main fabric	seam allowance 0.6 cm included; Sleeve hem: 1.0 cm
Neck binding (bias)	cut 1	Main fabric	seam allowance 0.4 cm included
Skirt front	cut 1 on fold	Main fabric	seam allowance 0.6 cm included; Hem: 1.0 cm
Skirt back	cut 2 (mirrored pair)	Main fabric	seam allowance 0.6 cm included; Hem: 1.0 cm; Back opening (turn under): 1.5 cm

Fabric needed

- **Main fabric:** about 0.25 m at 150 cm width (single layer, incl. 10 % for shrinkage; folded would need 0.35 m)
- Doll clothes: narrow seam allowances (6 mm). Sew with a short stitch (1.5–2 mm) and finish woven edges with pinking shears, a narrow zigzag or a little fray stop.
- Thin, soft fabrics work best (cotton poplin, batiste, old T-shirts); thick fabric makes the tiny seams bulky.
- The back opens completely: close it with narrow Velcro (1 cm) or 2–3 small snaps.

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

Cut on fold (Front, Skirt front): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

Validation report

Sewability check

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	3.1 cm	3.1 cm	+0.0 mm	□ OK	
Seam pairs	Armhole ↔ sleeve cap (ease)	ease 3–8 mm	13.8 cm ↔ 14.3 cm	+5.4 mm	□ OK	Woven: 3–8 mm
Seam pairs	Shoulder seam front ↔ back	3.7 cm	3.7 cm	+0.0 mm	□ OK	
Seam pairs	Sleeve seam (front ↔ back)	1.2 cm	1.2 cm	+0.0 mm	□ OK	
Seam pairs	Neck binding ↔ neckline	20.4 cm	20.4 cm	+0.0 mm	□ OK	
Seam pairs	Waist seam bodice ↔ skirt (gathered)	40–100 % of 60.9 cm	38.1 cm = 63 %	-228.4 mm	□ OK	gather ratio 1.6 (allowed 1–2.5)
Seam pairs	Skirt side seam front ↔ back	11.4 cm	11.4 cm	+0.0 mm	□ OK	
Seam pairs	Skirt centre back seam (left ↔ right)	7.4 cm	7.4 cm	+0.0 mm	□ OK	open for the top 4.0 cm (notch)
Angles/transitions	Underarm: side seam ↔ armhole front	90°	91.9°	+1.9°	□ OK	
Angles/transitions	Underarm: side seam ↔ armhole back	90°	92.2°	+2.2°	□ OK	

Area	Check	Target	Actual	Δ	Result	Note
Angles/transitions	Shoulder ↔ armhole (front + back)	180°	180.6°	+0.6°	✅ OK	
Angles/transitions	Sleeve cap top (smooth)	180°	178.2°	-1.8°	✅ OK	
Angles/transitions	Sleeve underarm ↔ cap front	90°	90.9°	+0.9°	✅ OK	
Angles/transitions	Shoulder ↔ neckline (front + back)	180°	180.9°	+0.9°	✅ OK	
Angles/transitions	Neckline at centre front	90°	90.4°	+0.4°	✅ OK	
Angles/transitions	Neckline at centre back	90°	89.0°	-1.0°	✅ OK	
Angles/transitions	Hem at centre front	90°	90.0°	+0.0°	✅ OK	
Angles/transitions	Hem ↔ side seam front	90°	88.8°	-1.2°	✅ OK	
Angles/transitions	Skirt hem ↔ side seam	90°	90.0°	+0.0°	✅ OK	
Angles/transitions	Skirt side seam ↔ waist	90°	90.0°	+0.0°	✅ OK	
Angles/transitions	Skirt waist ↔ centre	90°	90.0°	+0.0°	✅ OK	
Measurements	Chest circumference pattern = body + ease	36.6 cm	36.9 cm	+0.8 %	✅ OK	body 30 cm + +22 % ease
Measurements	Head fits through the back opening	≥ 33.6 cm	38.1 cm	+45.0 mm	✅ OK	neckline + both edges of the back opening ≥ head circumference + 5 %
Measurements	Dress goes on through the back opening	≥ 32.6 cm	46.0 cm	+134.9 mm	✅ OK	waist seam + both edges of the skirt opening ≥ hip + 5 %
Measurements	Dress length plausible	14.9 cm – 33.5 cm	20.7 cm	+0.0 mm	✅ OK	from the neck point
Geometry	Front: outline & seam allowance	closed, SA present	ok		✅ OK	seam allowance 0.6 cm included; Neckline: 0.4 cm
Labels	Front: grainline & cutting	present	cut 1 on fold		✅ OK	Main fabric
Geometry	Back: outline & seam allowance	closed, SA present	ok		✅ OK	seam allowance 0.6 cm included; Back opening (turn under): 1.5 cm; Neckline: 0.4 cm
Labels	Back: grainline & cutting	present	cut 2 (mirrored pair)		✅ OK	Main fabric
Geometry	Sleeve: outline & seam allowance	closed, SA present	ok		✅ OK	seam allowance 0.6 cm included; Sleeve hem: 1.0 cm
Labels	Sleeve: grainline & cutting	present	cut 2 (mirrored pair)		✅ OK	Main fabric
Geometry	Neck binding (bias): outline & seam allowance	closed, SA present	ok		✅ OK	seam allowance 0.4 cm included
Labels	Neck binding (bias): grainline & cutting	present	cut 1		✅ OK	Main fabric
Geometry	Skirt front: outline & seam allowance	closed, SA present	ok		✅ OK	seam allowance 0.6 cm included; Hem: 1.0 cm

Area	Check	Target	Actual	Δ	Result	Note
Labels	Skirt front: grainline & cutting	present	cut 1 on fold		□ OK	Main fabric
Geometry	Skirt back: outline & seam allowance	closed, SA present	ok		□ OK	seam allowance 0.6 cm included; Hem: 1.0 cm; Back opening (turn under): 1.5 cm
Labels	Skirt back: grainline & cutting	present	cut 2 (mirrored pair)		□ OK	Main fabric
Geometry	Back: turned allowance lies flat	≤ 0.5 mm	0.0 mm		□ OK	Back opening (turn under): corners mirrored at the fold line
Geometry	Sleeve: turned allowance lies flat	≤ 0.5 mm	0.0 mm		□ OK	Sleeve hem: corners mirrored at the fold line
Geometry	Front: cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.4° off 90°		□ OK	Centre (fold)
Geometry	Skirt front: cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.0° off 90°		□ OK	Centre (fold)
Labels	Front: notches	on the cutting line	1 notch mark(s)		□ OK	
Labels	Back: notches	on the cutting line	2 notch mark(s)		□ OK	
Labels	Sleeve: notches	on the cutting line	4 notch mark(s)		□ OK	1 at corners
Labels	Skirt back: notches	on the cutting line	2 notch mark(s)		□ OK	

Result: 45 OK, 0 warning(s), 0 error(s): the pattern is sewable.

Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"chest_width": 366}}`).

Name	Value	Derived from	Note
`chest_width`	36.6 cm	chest × (1 + 22 % ease)	
`neck_width_half`	3.5 cm	neck / 5 + 1.5 cm	
`front_neck_depth`	4.0 cm	neck_width_half + 1.5 cm	
`back_neck_depth`	0.5 cm	fixed	
`shoulder_length`	3.7 cm	shoulder + allowance for fit 'wide'	
`shoulder_slope`	17	fixed	
`armhole_depth`	6.3 cm	18 % of chest_width + 2.5 cm + 3 %, limited to option armhole_depth_max	
`body_length`	9.3 cm	option body_length	

ting layout · Dress with gathered skirt doll 45 cm (made to measu

Main fabric: 150 cm wide, single layer
needed ≈ 0.25 m

